

UTILIZING VLM FOR EXTRACTING PHOTOS OF NAMED INDIVIDUALS FROM HISTORICAL NEWSPAPERS

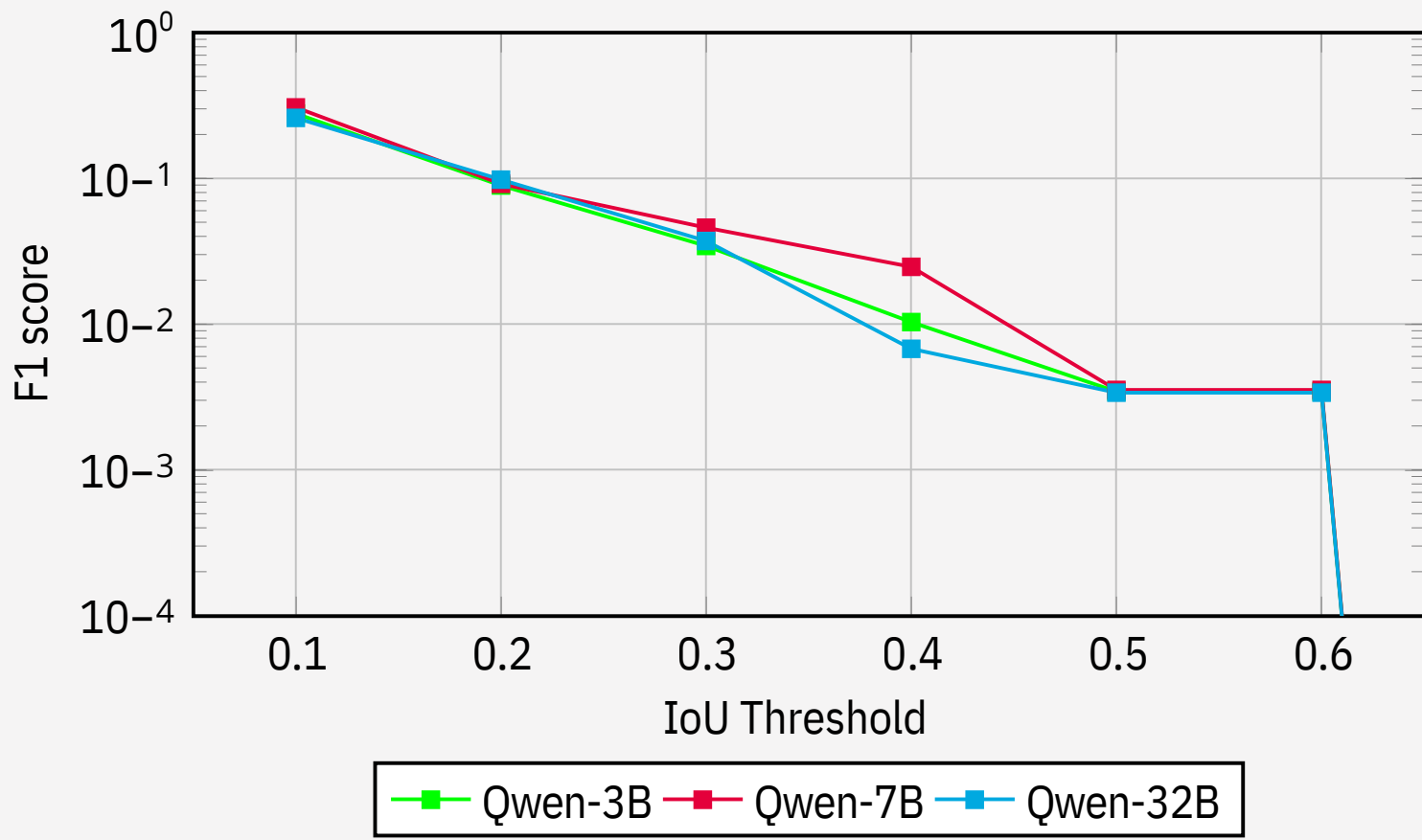
Authors: Petr Bařcar, Lukáš Katona, Jakub Vodrážka
Supervisor: Ing. Marek Vaško

Problem Description

Query: Models were provided with newspaper scan and target person’s name
Task: Find bounding box of target person’s portrait

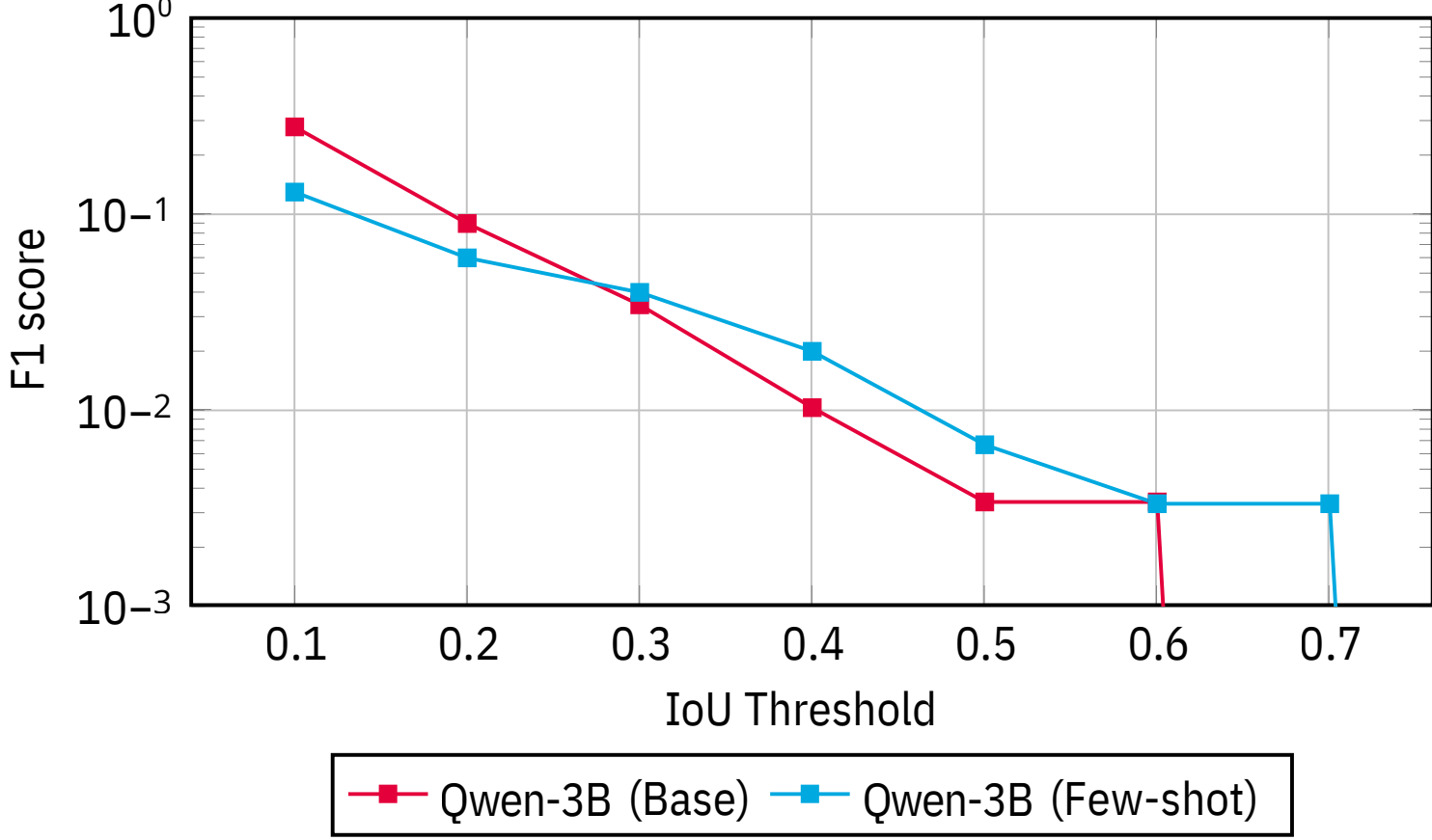
Baseline Tests

F1 Comparison of base performance of models:
Qwen2.5-VL-Instruct (3B), Qwen2.5-VL-Instruct (7B), Qwen2.5-VL-Instruct (32B)



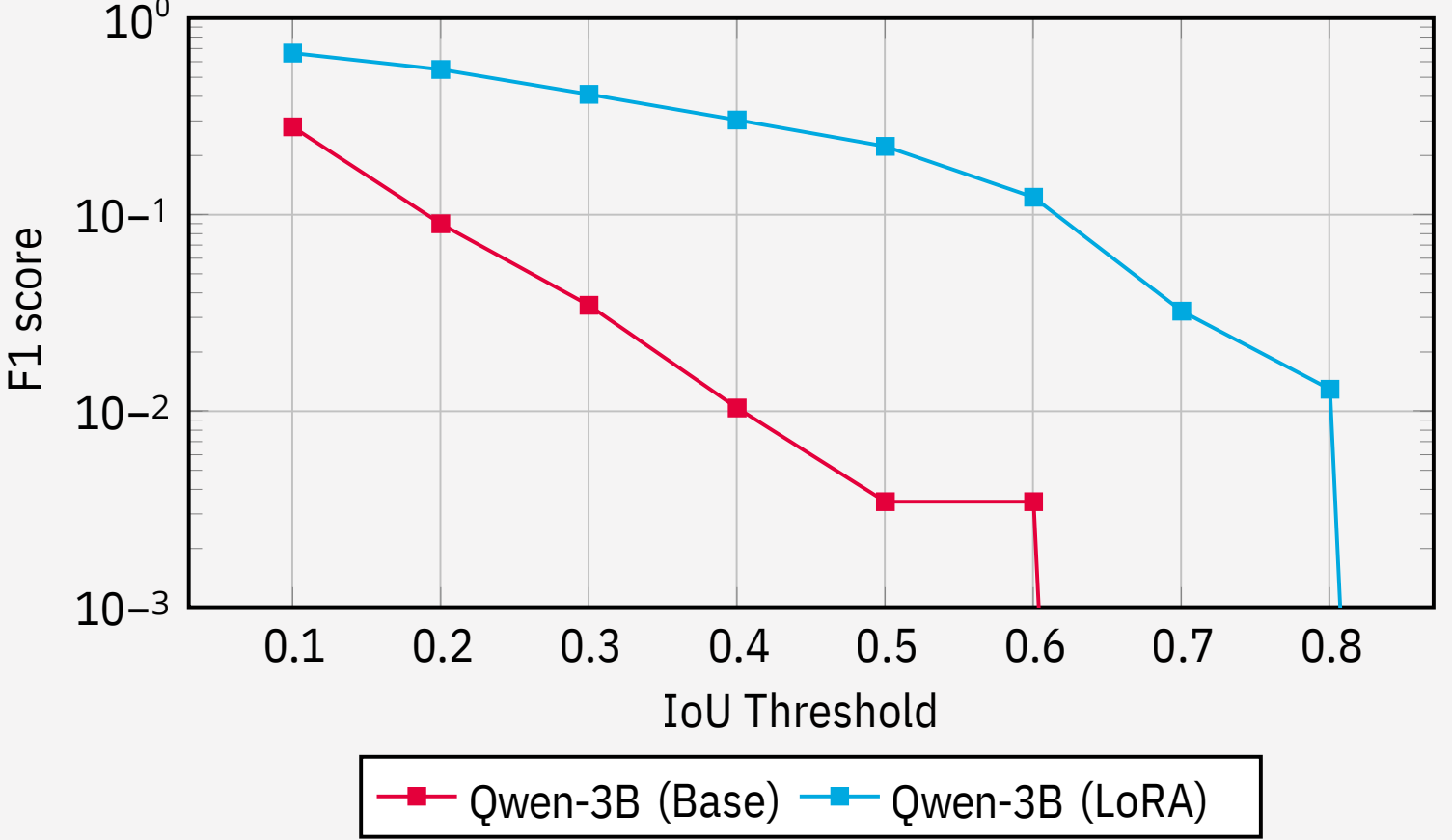
Few-Shot Learning

F1 Comparison of few-shot and baseline model
Qwen2.5-VL-Instruct (3B)



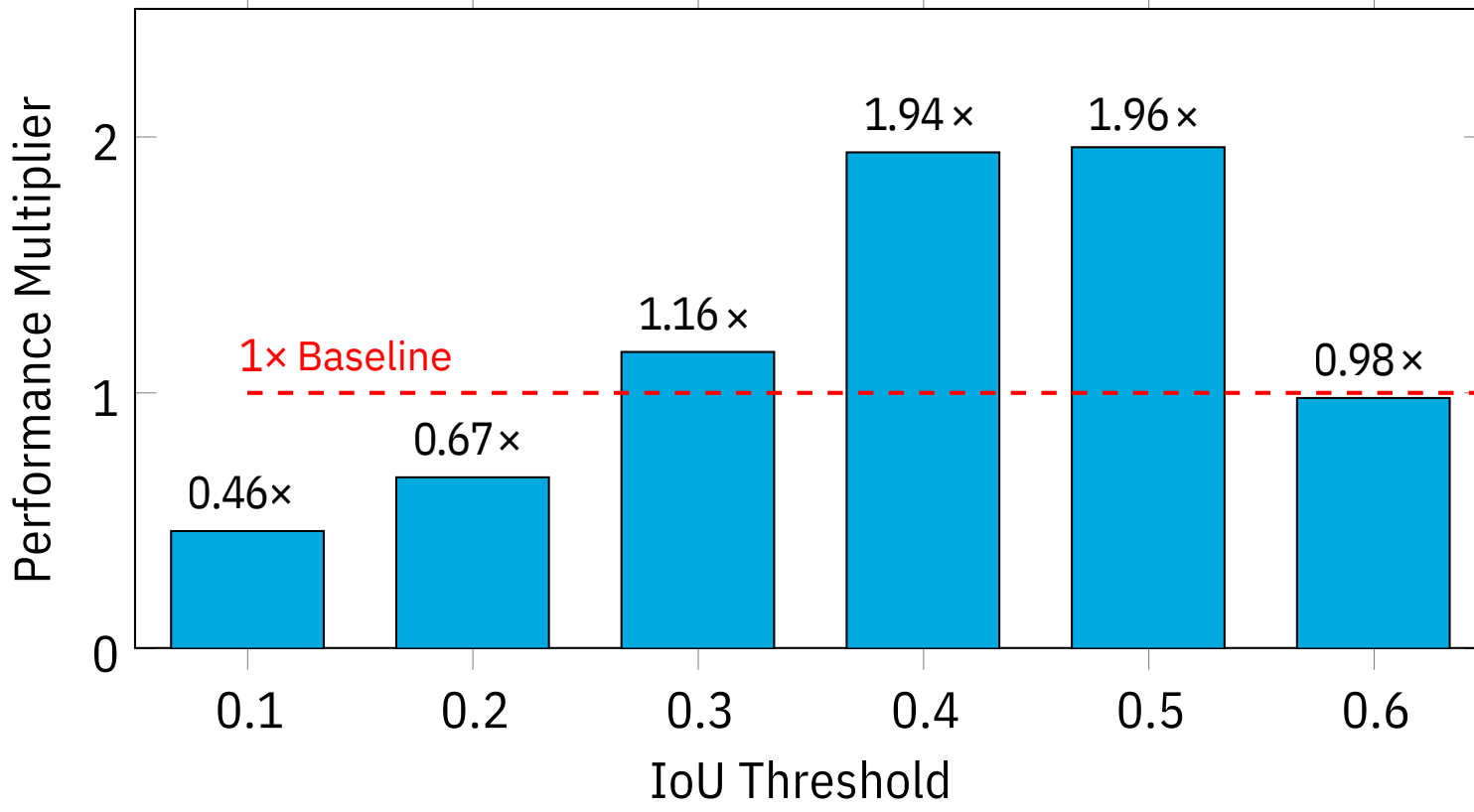
LoRA Finetuning

F1 Comparison of finetuned LoRA and baseline model
Qwen2.5-VL-Instruct (3B)



Performance Multiplier:

Qwen2.5-VL-Instruct (3B) (Few-Shot)



Performance Multiplier:

Qwen2.5-VL-Instruct (3B) (LoRA)

